

Chit 1 — MySQL: Joins and Subqueries

Table Creation & Data Insertion

```
-- Create Customer Table
CREATE TABLE Customer (
  C_Id INT PRIMARY KEY,
  Cname VARCHAR(50),
  City VARCHAR(50)
);

INSERT INTO Customer VALUES
(1, 'John', 'Nashik'), (2, 'Seema', 'Aurangabad'),
(3, 'Amita', 'Nagar'), (4, 'Rakesh', 'Pune'),
(5, 'Samata', 'Nashik'), (6, 'Ankita', 'Chandwad'),
(7, 'Bhavika', 'Pune'), (8, 'Deepa', 'Mumbai'),
(9, 'Nitin', 'Nagpur'), (10, 'Pooja', 'Pune');

-- Create Account Table
CREATE TABLE Account (
  C_Id INT,
  Acc_Type VARCHAR(20),
  Amount DECIMAL(10,2),
  FOREIGN KEY (C_Id) REFERENCES Customer(C_Id)
);

INSERT INTO Account VALUES
(1, 'Current', 5000), (2, 'Saving', 20000),
(3, 'Saving', 70000), (4, 'Saving', 50000),
(6, 'Current', 35000), (7, 'Loan', 30000),
(8, 'Saving', 50000), (9, 'Saving', 90000),
(10, 'Loan', 8000), (11, 'Current', 45000);
```

Q1. Show cname, Acc_Type, amount of customers having Saving account

```
SELECT c.Cname, a.Acc_Type, a.Amount
FROM Customer c
JOIN Account a ON c.C_Id = a.C_Id
WHERE a.Acc_Type = 'Saving';
```

Q2. Display data using Natural, Left and Right Join

```
-- NATURAL JOIN
SELECT * FROM Customer NATURAL JOIN Account;

-- LEFT JOIN
SELECT c.C_Id, c.Cname, c.City, a.Acc_Type, a.Amount
FROM Customer c
LEFT JOIN Account a ON c.C_Id = a.C_Id;

-- RIGHT JOIN
SELECT c.C_Id, c.Cname, c.City, a.Acc_Type, a.Amount
FROM Customer c
RIGHT JOIN Account a ON c.C_Id = a.C_Id;
```

Q3. Display information of customers living in same city as 'Pooja'

```
SELECT * FROM Customer
```

```
WHERE City = (  
    SELECT City FROM Customer WHERE Cname = 'Pooja'  
);
```

Q4. Display accounts having amount less than average amount

```
SELECT * FROM Account  
WHERE Amount < (SELECT AVG(Amount) FROM Account);
```

Q5. Display C_id having maximum amount

```
SELECT C_Id, Amount FROM Account  
WHERE Amount = (SELECT MAX(Amount) FROM Account);
```

Q6. Display amount and acc_type where amount is minimum of that Acc_Type

```
SELECT Acc_Type, Amount FROM Account a  
WHERE Amount = (  
    SELECT MIN(Amount) FROM Account  
    WHERE Acc_Type = a.Acc_Type  
);
```

Q7. Display accounts with amount higher than any saving account amount

```
SELECT * FROM Account  
WHERE Amount > ANY (  
    SELECT Amount FROM Account WHERE Acc_Type = 'Saving'  
);
```